

Tuberculosis of the Urinary Tract

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DEFINITION

More than a century after *Mycobacterium tuberculosis* (MTB) was first described by Robert Koch, tuberculosis (TB) remains a global health problem and among the leading causes of death from infectious disease worldwide.¹ As per the World Health Organization (WHO) Global Tuberculosis Report in 2018, an estimated 10 million people developed TB, of whom 4 million people with TB remained undiagnosed and untreated.² Infection with MTB is responsible for nearly 1.3 million deaths each year. TB is the most common opportunistic infection in people with human immunodeficiency virus (HIV).³ In HIV-endemic areas, as many as 75% of patients with urogenital TB (UG-TB) are coinfecting with HIV. TB is also common in patients with chronic kidney disease (CKD). In some endemic areas, TB has been reported to occur in up to 8.7% of patients on hemodialysis, 12.3% of kidney allograft recipients,⁴ and 9.3% of children with nephrotic syndrome.⁵ Although the global incidence is estimated to be declining by 1.6% and mortality by 4.1% per year, few countries are likely to meet the United Nations (UN) Sustainable Development Goals target to end the epidemic by 2030.

Drug-resistant forms of tuberculosis are currently among the world's deadliest pathogens. Data from 2017 show that among nearly 0.55 million cases of rifampicin-resistant TB, 80% had multidrug-resistant (MDR) TB (defined as resistance to rifampicin and isoniazid) and 8.5% of patients with MDR-TB had extensively drug-resistant

(XDR) TB (with XDR-TB defined as resistance to rifampicin, isoniazid, quinolones, and aminoglycosides).⁶

The common sites of extrapulmonary involvement by MTB are lymph nodes, urinary tract, genital tract, bone, adrenal glands, and the central nervous system. UG-TB may remain asymptomatic and undiagnosed in up to 20% of individuals with active pulmonary TB. Kidney involvement may also occur as part of miliary TB. Because UG-TB can easily be overlooked, a high index of suspicion is necessary for early diagnosis in endemic areas. Delayed diagnosis may result in irreversible organ damage and even end-stage kidney failure.⁷

ETIOLOGY

TB is caused by the *M. tuberculosis* complex (MTBC). *Mycobacterium tuberculosis* is a nonmotile, nonsporing, rod-shaped, slightly curved, obligate aerobic bacterium (Fig. 54.1). It is weakly gram-positive and acid- and alcohol-fast. The cell wall has a high content of mycolic acid that makes the organism resistant to acids, alkalis, antibiotics, osmotic agents, and intracellular events. Lipoarabinomannan (LAM) forms a part of the mycolic acid in the cell wall and its identification in urine is useful for diagnosis. The most superficial layer consists of glycolipids and polypeptide (Fig. 54.2). The polypeptides are isolated and purified to make the skin testing antigen, which is called purified protein derivative (PPD) (Table 54.1). Although most TB, including genitourinary TB, is caused by MTB, other mycobacteria, such as *M. avium-intracellulare*, *M. kansasii*, *M. bovis*, *M. fortuitum*, and *M. szulgai*, may cause clinical disease, especially in immune-compromised hosts. Among the atypical mycobacteria, *M. chelonae abscessus* belongs to the group of "rapid growers" and causes abscesses after injection or percutaneous procedures, especially in patients with diabetes.⁸

PATHOGENESIS

The most common routes of transmission are:

1. Inhalation of infected droplets from patients with active pulmonary TB (95%)
2. Ingestion of dairy products infected with *M. bovis* from cattle
3. After therapeutic intravesical bacillus Calmette-Guérin (BCG) instillation
4. Contiguous spread to nearby organs
5. Lymphatic or hematogenous spread
6. Congenital, sexual, or accidental inoculation (rare)

The clinical and pathologic manifestations of TB depend on the virulence of the organism and the effectiveness of the host response. In 95% of people, after primary infection, the bacilli multiply locally and evoke a local response to form a primary granuloma called the Ghon complex. The immune responses result in either eradication, containment as latent TB infection (LTBI), or active disease. The WHO has defined LTBI

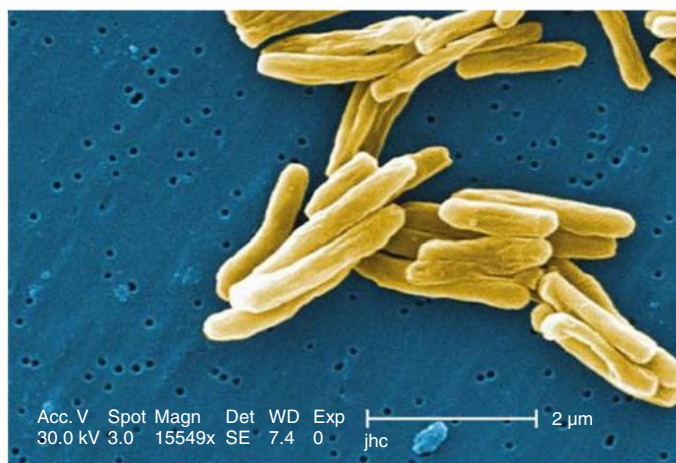


Fig. 54.1 Scanning Electron Microscopy of *Mycobacterium Tuberculosis*. The bacillus is a straight or slightly curved rod about 2 to 4 μm long and 0.3 to 0.5 μm wide. (High magnification $\times 15,549$). (Courtesy Ray Butler MS, CDC. Division of Tuberculosis Elimination, National Center for HIV, STD, and TB Prevention. Atlanta, GA.)

Diagrammatic Representation of Mycobacterial Cell Wall

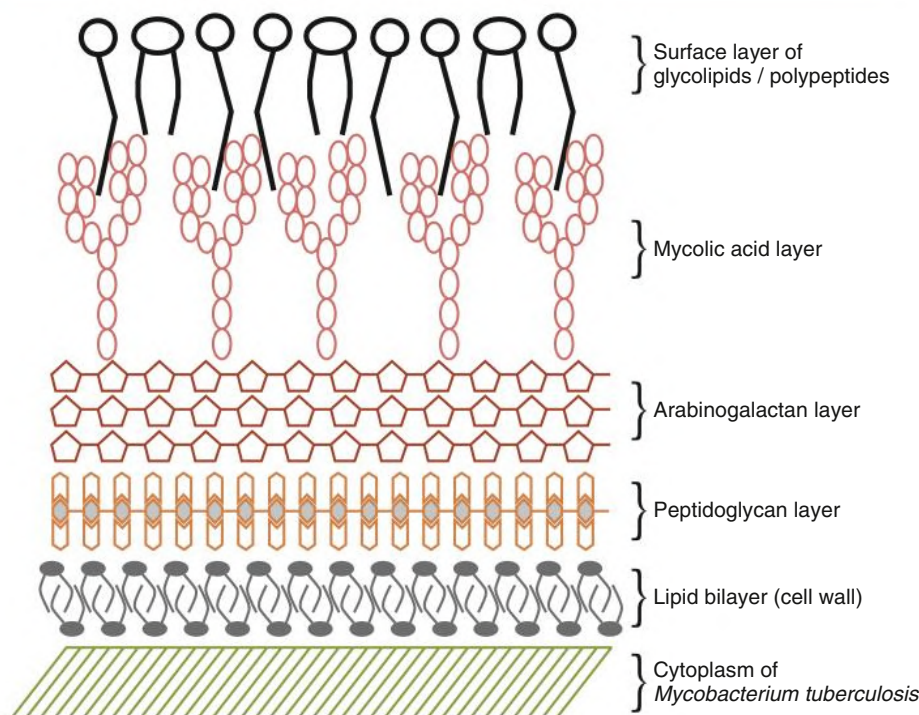


Fig. 54.2 Diagrammatic representation of mycobacterial cell wall.

TABLE 54.1 Cell Wall Characteristics Affecting Survival of *Mycobacterium tuberculosis*

Composition of Cell Wall	Factors for Survival
Mycolic acid	Resist proteolysis and uptake into phagolysosomes
Lipoarabinomannan	Detected by CD1 T cells that may assist the immune response
Muramyl dipeptide	Stimulate T cell responses (granuloma formation) that may assist clearance
Glycolipids	Inhibit macrophage function
Inert lipids and surface proteins	Dormant survival inside phagocytes

as a state of persistent immune response to stimulation by MTB antigens with no evidence of clinically manifest active TB. The Ghon complex may be found in the lungs, tonsils, gut, or any organ. The organisms from the primary focus drain to a regional lymph node or nodes, causing their enlargement. This triad of the primary Ghon focus, lymphangitis, and lymphadenitis is known as the primary complex and is self-limited in 90% to 95% of cases. Progression occurs in the remainder, leading to local or systemic dissemination of mycobacteria via lymphatics or the bloodstream.⁹ Because of the slow replication rate, its intracellular location in macrophages, and acquired immune responses, it may take 1 to 2 years after primary infection for symptoms and signs to manifest. Reactivation of LTBI may occur in 5% to 15% of people and progress to active TB. Up to 10% of patients with UG TB have active pulmonary TB, and 50% have evidence of previous TB on chest radiographs.

Bacilli from the regional lymph node enter the bloodstream through the thoracic duct resulting in silent dissemination to various sites, including the kidney cortex (Fig. 54.3). Here, the bacilli elicit an inflammatory

response, resulting in granuloma formation that may heal and form a microscopic scar, remain dormant for many years, or rupture into the proximal tubule of the nephron. The bacilli in the nephron are trapped at the level of the loop of Henle, where they multiply. The relatively poor blood flow, hypertonicity, and high ammonia concentration in the kidney medulla impair the immune response and favor the formation of medullary granulomas, leading to chronic tubulointerstitial nephritis, fibrosis, or papillary necrosis. The center of these granulomas (tuberculous), which contains macrophages, may undergo coagulative necrosis and form ulcers or cavities filled with cheese-like caseous material (Fig. 54.4A). Extensive caseous destruction of the kidney parenchyma occurs with formation of lobules and cavities (see Fig. 54.4B). The dissemination of infection to the kidney pelvis can cause tuberculous pyelonephritis. This may burst into perirenal tissue to form perinephric abscess or into the calyx causing pyonephrosis. With progressive fibrosis and scarring of the uretero-pelvic junction, dilation of pelvis and calyces may occur. Extensive necrosis replaces the kidney parenchyma with caseous material. Various patterns of calcification may be seen in imaging (Fig. 54.5). Calcification commences intracellularly because of the accumulation of phosphate ions from the disintegration of nucleoproteins and entry of calcium ions from cell membrane damage. These dystrophic calcific lesions may harbor live or dormant mycobacteria and such dystrophic lesions are considered active disease and not signs of healing. Dystrophic calcification may result in a nonfunctioning kidney (autonephrectomy) with dystrophic calcification called *putty* or *cement* kidney. Kidney TB can be briefly classified as:

- Stage 1: nondestructive form (TB of parenchyma)
 - Stage 2: small-destructive form (TB papillitis)
 - Stage 3: destructive form with one or two caverns (cavernous kidney TB)
 - Stage 4: widespread destructive form (polycavernous kidney TB)
- Involvement of the ureter is common and frequently affects the lower third and ureteropelvic junction leading to obstructive uropathy.

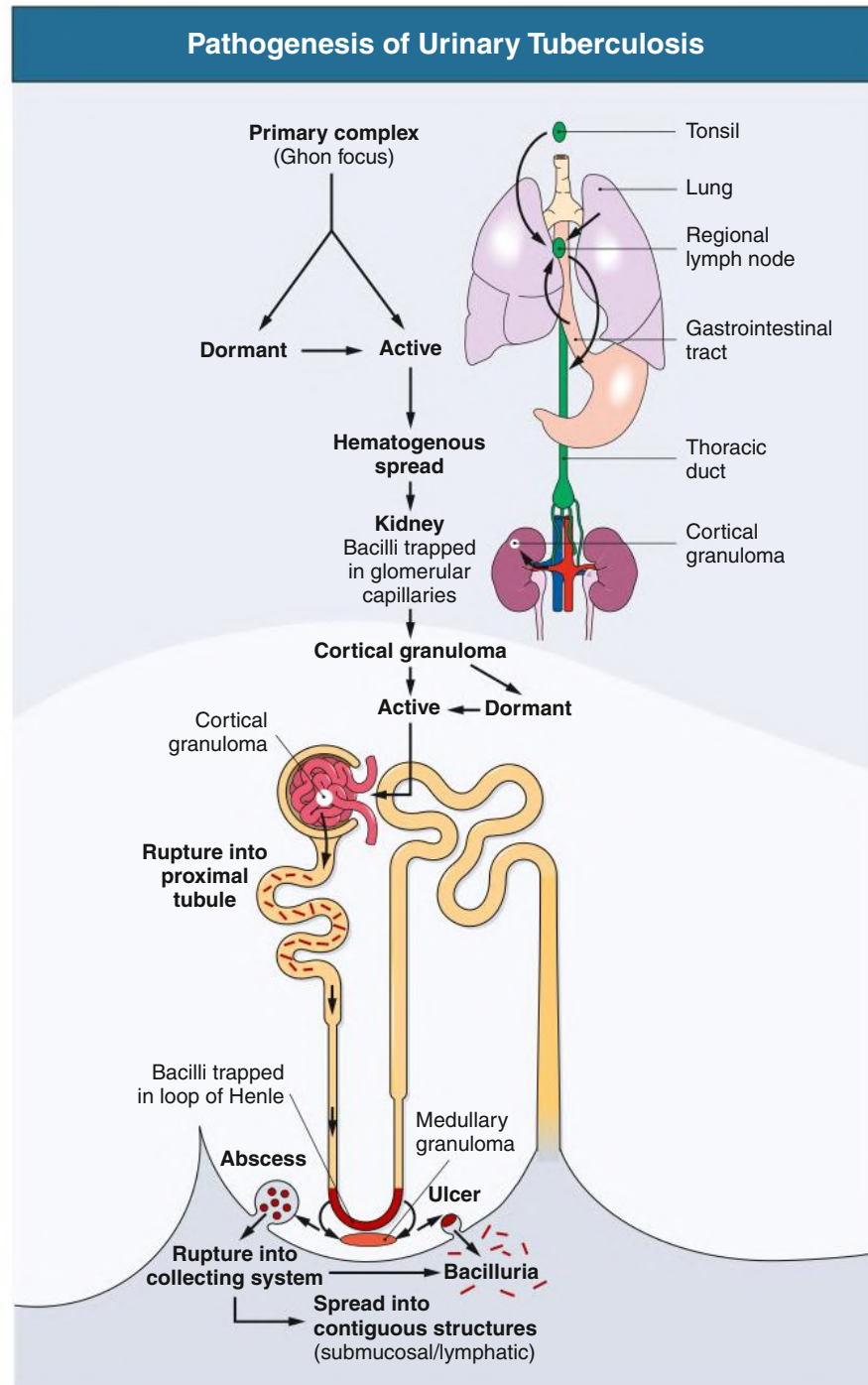


Fig. 54.3 Pathogenesis of urinary tuberculosis.

Ureteric involvement leads to inflammation, edema, granulomatous ulceration, and fibrosis, leading to irregular ureteric strictures (Fig. 54.6A), segmental dilation (“corkscrew” ureter), or fibrosis resulting in a shortened and rigid ureter (“pipe-stem” ureter).

The bladder may develop hyperemia near the ureteral orifice, followed by superficial ulcers and granulomatous changes involving all layers (pancystitis). Healing by fibrosis at the ureteral orifice results in a refluxing “golf-hole” ureter. Extensive fibrosis of the bladder wall results in a thick, small-capacity bladder called a “thimble” bladder (see Fig. 54.6B). Bladder infection also may rarely result from instillation of BCG in the bladder as part of treatment of superficial bladder carcinomas. Rare complications of bladder TB include vesicovaginal,

vesicocolic, and enterovesical fistulae and bladder perforation. TB of the bladder can be classified into four stages¹⁰:

- Stage 1: infiltrative bladder TB (tubercles in bladder wall)
- Stage 2: ulcerous bladder TB (erosion of bladder mucosa)
- Stage 3: interstitial cystitis (painful bladder syndrome)
- Stage 4: contracted bladder (obliteration)

Involvement of the genital tract is also common. Men with TB of the urinary tract have epididymitis, prostatitis, seminal vesiculitis, orchitis, cold abscesses, or discharging scrotal sinuses. In women, genital tract involvement is less common, but, if present, usually presents as salpingitis, often diagnosed during investigation for infertility.

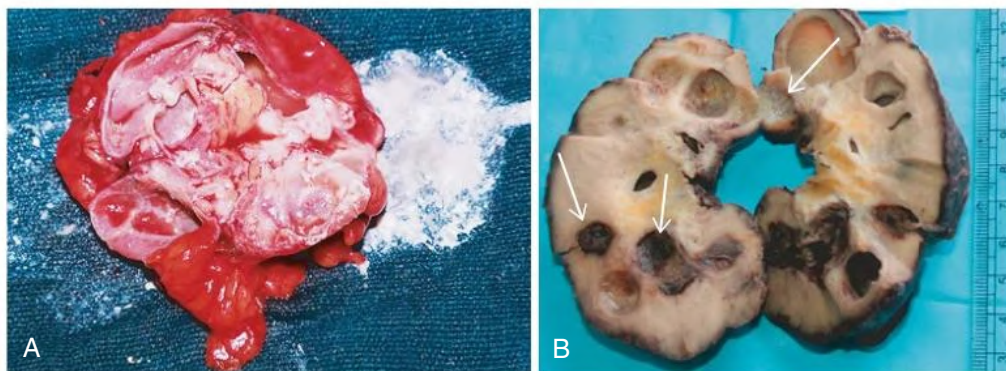


Fig. 54.4 Kidney Tuberculosis (Cut Section). (A) Cut section of kidney shows areas of cavitation filled with white chalky material (caseation necrosis). (B) Cavitating lesions (*arrows*) caused by tuberculosis in cut section of kidney (autopsy). (Courtesy Department of Pathology, Amrita Institute of Medical Sciences, Kochi, Kerala, India.)

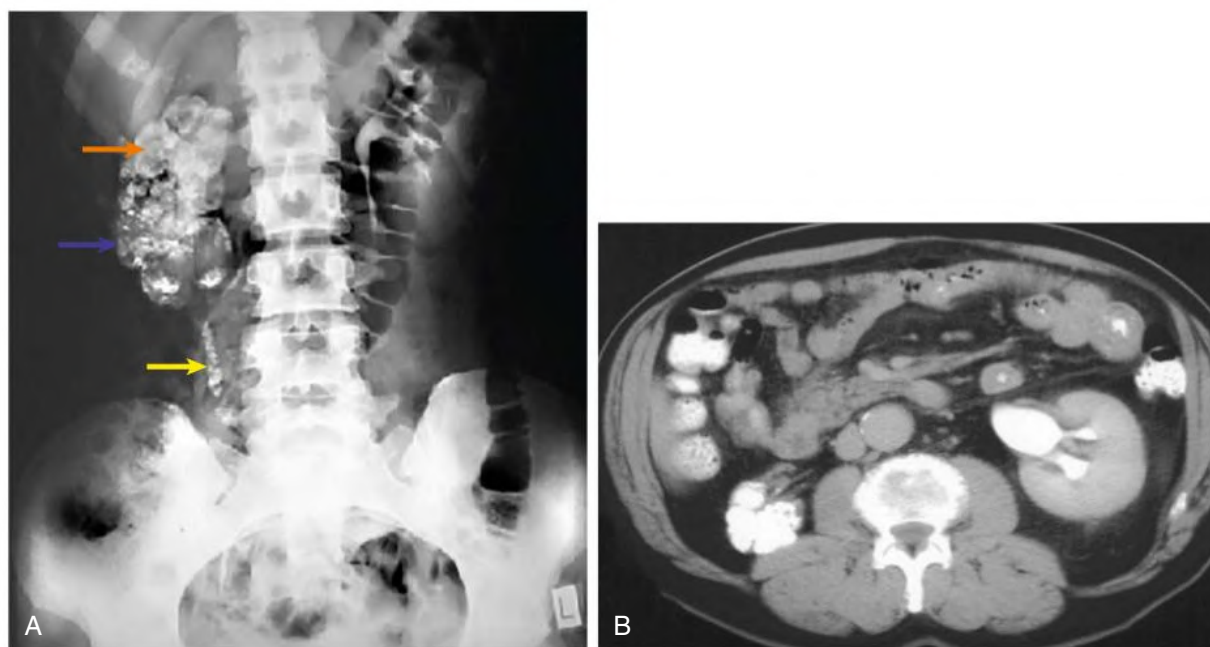


Fig. 54.5 Types of calcification. (A) Curvilinear pattern (*orange arrow*), amorphous and speckled pattern (*blue arrow*), ureteric calcification (*yellow arrow*). (B) Contrast-enhanced computed tomography image showing contracted densely calcified right kidney and normally functioning left side (density on right side was the same before and after contrast).

TB can be transmitted from the donor to recipient through kidney transplantation, and the risk can be minimized by thorough evaluation of the donor.¹¹

CLINICAL MANIFESTATIONS

UG-TB can present at any age, but most patients are 20 to 40 years of age, with a male-to-female ratio of 2:1. It is relatively uncommon in children because of the long latency period. It may be asymptomatic and detected incidentally during investigations for other illnesses. Clinical presentation may include a range of manifestations (Table 54.2), but a high index of suspicion enables early diagnosis. Risk factors for TB include one or more of the following:

1. Close contact with sputum-positive individuals
2. Residence in an endemic area
3. Overcrowding/social deprivation/vagrancy
4. Immunosuppression (HIV/infection/diabetes mellitus)

5. CKD
6. Vitamin D deficiency
7. Malnutrition and other debilitating illnesses

Asymptomatic Presentation

In some countries, one-quarter of the population may have latent MTB.¹¹ Almost 25% of patients have no clinical or laboratory evidence of any abnormality, and the diagnosis of urinary TB is made on investigation for other diseases, during surgery, or at autopsy. Another 25% may have asymptomatic urinary abnormalities, particularly pyuria or hematuria. Nonspecific constitutional symptoms like fever, weight loss, and night sweats are uncommon. UG-TB should be suspected when conventional urine cultures are repeatedly negative in spite of empiric antibiotic therapy for suspected UTI. The urine pH may be acidic with persistent pyuria and negative routine bacterial cultures (acid-sterile pyuria). Genital involvement in males may be with chronic epididymitis or prostatitis that does not resolve with standard antibiotics.¹¹

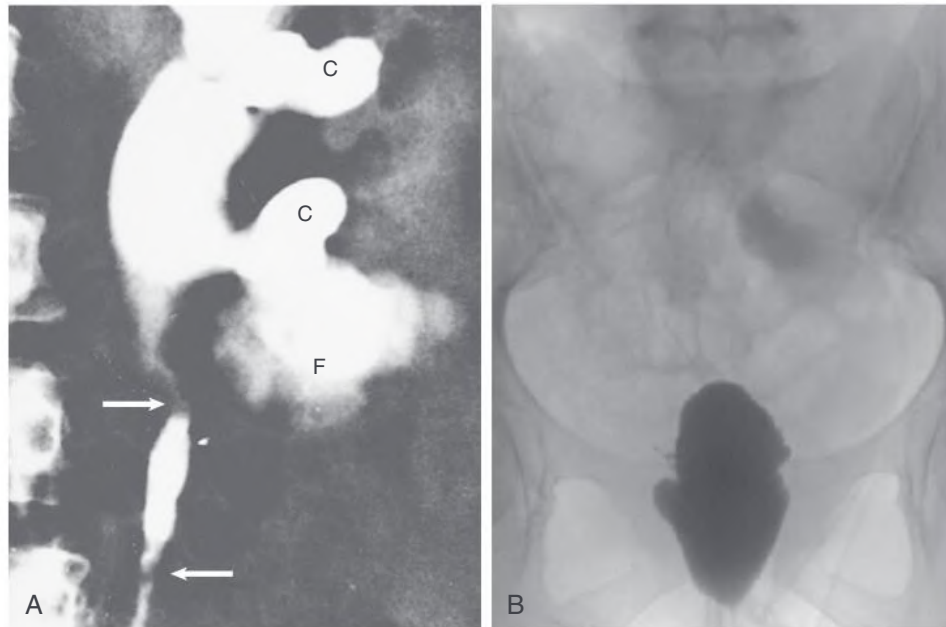


Fig. 54.6 Ureteric and Bladder Changes. (A) Multiple ureteric strictures (arrows), caliectasis (C), and fuzzy calyx (F). (B) Cystogram showing irregular, small capacity bladder or "thimble bladder."

TABLE 54.2 Clinical Features of Urinary Tuberculosis

Features	Frequency (%)	Symptoms
Asymptomatic	25	Detected during autopsy, surgery, or investigations for other diseases
Asymptomatic urinary abnormalities	25	Persistent pyuria, microscopic abnormalities, hematuria
Lower urinary tract symptoms (most common)	40	Frequency, urgency, dysuria, incontinence, nocturia, suprapubic pain, perineal pain
Male genital tract involvement	75	Epididymitis, hemospermia, infertility, reduced semen volume
Female genital tract involvement	<5	Amenorrhea, infertility, vaginal bleeding, pelvic pain
Constitutional symptoms	<20	Fever, reduced appetite, anorexia, weight loss, night sweats
Miscellaneous	—	Urolithiasis, hypertension, acute kidney injury, chronic kidney disease, abdominal colic, abdominal mass

Lower Urinary Tract Symptoms

Of the patients who are symptomatic, lower urinary tract symptoms, such as frequency, urgency, dysuria, nocturia, frank pyuria, or hematuria, occur in more than 75% of patients. Increased urinary frequency is the earliest and most common symptom. Recurrent bouts of painless gross macroscopic hematuria should alert the clinician to the possible diagnosis of urinary TB, although glomerular diseases such as immunoglobulin A (IgA) nephropathy also should be considered. Macroscopic

hematuria in urinary TB is a result of bleeding from the ulcerating lesions, inflammation of the urothelium, or rupture of a blood vessel in the vicinity of a cavity. Episodes of frank pyuria also may be a manifestation of kidney TB and indicate either secondary bacterial infection or drainage of a caseous focus into the collecting system. In advanced disease, frequency and urgency of micturition related to reduced bladder capacity, incomplete emptying, increased susceptibility to infection, and secondary vesicoureteral reflux (VUR) may also occur.

Kidney Tuberculosis

Patients with kidney TB may be asymptomatic with only abnormal urinalysis. With advancing kidney TB, the clinical presentations are nonspecific symptoms and signs such as lower urinary symptoms, flank pain, colic, and hematuria. Long-standing kidney parenchymal involvement may result in tubular proteinuria, which is often in the subnephrotic range. The kidneys are of normal size and show diffuse interstitial nephritis, with noncaseating or caseating granulomas. MTB can be identified by molecular tests in nearly 75% of biopsy samples. The overall incidence of kidney failure reported in the literature is 24%. The mechanisms of kidney failure are¹²:

- Kidney parenchymal infection causing obliterative endarteritis
- Extensive dystrophic calcification
- Secondary kidney amyloidosis
- Multiple ureteric strictures with postobstructive atrophy
- Tubulointerstitial nephritis
- Destruction of kidney parenchyma by caseation

Nephrotic proteinuria occurs rarely, and it could be because of mesangioproliferative glomerulonephritis or serum amyloid A (SAA) protein-associated secondary amyloidosis. Chronic inflammation of the kidney pelvis can also lead to squamous metaplasia or carcinoma.

Ureteral Tuberculosis

Nonspecific manifestations like hematuria or abdominal colic occur because of ureteric obstruction associated with kidney stones, blood clots, or sloughed kidney papillae. Secondary infection with *Escherichia coli* is also common.

Genital Involvement

In males, the most common genital involvement is epididymitis, which manifests with scrotal discomfort, mass, or cold abscess or may be asymptomatic. When the cold abscess ruptures, a nonhealing posterior scrotal sinus discharging caseous material may form. Thickening of the vas deferens may result in a “beaded” texture. TB of the prostate may manifest with lower urinary tract symptoms and perineal pain. The prostate may be hard or boggy. Penile and urethral TB may manifest with strictures, fistulas, ulcers, or papulonecrotic skin lesions. Hemospermia, reduction of semen volume, and infertility are other manifestations of genital involvement. Direct spread of MTB to a sexual partner is also possible. Only 5% of females with kidney TB also have genital TB. The major manifestation of genital involvement in females is infertility resulting from adherent salpingitis. Secondary amenorrhea, vaginal bleeding, and pelvic pain caused by inflammation may occur.

Other Manifestations

Anemia is seen in less than 20% of patients with nonmiliary disease, but the frequency is higher in those with CKD stages 3 to 5. A few patients may develop nephrogenic diabetes insipidus or renal tubular acidosis. Hyporeninemic hypoaldosteronism may result from the tubulointerstitial injury secondary to obstructive uropathy.¹³ Hypertension is unusual in kidney TB, but intimal proliferation of vessels near inflamed areas may lead to segmental ischemia and renin release and hypertension.¹⁴ Relief of obstruction or removal of a nonfunctioning kidney causing hypertension may be necessary in rare instances.

Constitutional symptoms, such as fever, weight loss, night sweats, fatigue, and anorexia, occur in less than 20% of patients and indicate active infection in other organs or secondary bacterial infection of the urinary tract. All patients who present with constitutional symptoms must be carefully examined to identify pulmonary, lymph node, or skeletal TB. The chest radiograph may show evidence of active or healed tuberculous lesions in more than half of cases. Rarely, hypercalcemia may be present because of increased synthesis of calcitriol in the granulomas.

PATHOLOGY

The pathologic process of urinary TB may take the form of a miliary, granulomatous, or ulcerocavernous lesion depending on the virulence of the organism and host response. In the miliary form, which is seen particularly in immunosuppressed individuals, the cortex is studded with yellowish white, hard, pinhead-sized nodules that on microscopy show several coalescent granulomas with central caseation.

In early disease and in those with effective cell-mediated response, granulomas are seen in the kidney parenchyma, characterized by macrophages with engulfed bacilli, surrounded by epithelioid cells and Langhans giant cells (Fig. 54.7). Surrounding this, a cuff of lymphocytes and plasma cells are seen.¹⁵ Healing occurs by fibrosis and scarring.

In the more common ulcerocavernous form, yellow nodules consisting of pinhead-sized caseous foci are often seen on the outer surface. On cut section, granulomas and ulcers in the kidney pyramid or medullary cavities may be seen. Larger cavities filled with amorphous, soft, chalk-like eosinophilic material (caseous material) communicating with the collecting system also may be seen (see Fig. 54.2A). Other gross findings include multiple ulcers in the infundibular region of the calyces, calyceal stenosis, caliectasis (see Fig. 54.6A), ulcers, or strictures of the ureter with hydronephrosis, pyonephrosis, subcapsular collections, and perinephric abscesses. The bladder may show ulcers or be grossly fibrotic and contracted. In advanced cases, extensive caseating necrosis, replacing the normal kidney architecture, may be seen,

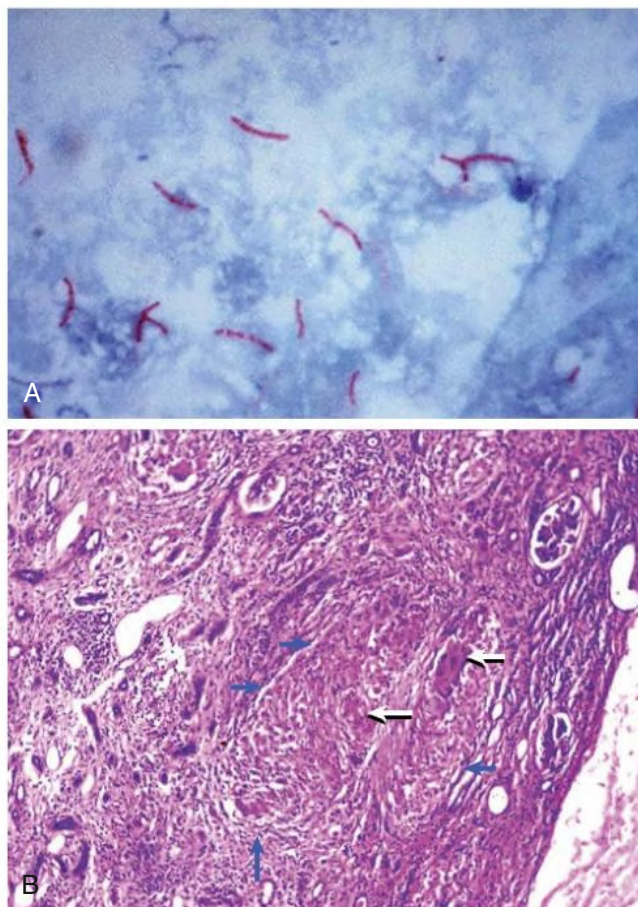


Fig. 54.7 Histologic Changes. (A) Acid fast bacilli in tissue (Ziehl Neelsen stain; $\times 1000$ magnification). (B) Needle biopsy of kidney showing granuloma formation (blue arrows) and multinucleated Langhans giant cells (white arrows). (A, Courtesy Dr. George Kubica, Centers for Disease Control and Prevention. B, Courtesy Department of Pathology, Medical College, Kottayam, India.)

which may undergo calcification. The presence of caseous necrosis or calcification suggests activity of the disease and is not to be taken as a sign of healing. Kidneys also may be enlarged from amyloidosis.

DIAGNOSIS

Early and accurate diagnosis of UG-TB is important for successful treatment outcomes, and a high index of suspicion is necessary. Elderly patients, as well as those exposed to or who have had infection in the past, immunocompromised individuals, and patients with TB elsewhere are at high risk. Although the gold standard for specific diagnosis of TB is by identifying MTB from a clinical sample, it may not be possible in all cases because of the paucibacillary nature of the disease. No single specific diagnostic test exists, and a range of microbiological, molecular, histopathologic, and imaging tests are available for the diagnosis.

Smear Microscopy

Microscopic examination of urine for acid-fast bacilli (AFB) using Ziehl-Neelsen (ZN) or auramine staining has been a first-line diagnostic test for UG-TB for the past 70 years. Bacterial load of 5000 organisms/mL is needed for positive smear and at least three early morning urine samples should be analyzed. Diagnostic yield from urine sample is only less than 40%. WHO recommends that conventional fluorescence

microscopy may be replaced by light-emitting diode (LED) microscopy.¹⁶ Contamination of urine with environmental, nonpathogenic mycobacteria such as *M. smegmatis* may give false-positive results.

Urine Culture

Culture of specimens for *MTB* is the gold standard for the diagnosis of active TB with a sensitivity of 65% and specificity of 100%. Traditional culture requires about 6 to 12 weeks for a result. The WHO has recommended automated liquid mycobacteria growth indicator tube (MGIT) culture as the gold standard where positive results can be available within 2 weeks.¹⁷ Because the organism is intermittently excreted, at least three, but preferably five, consecutive early morning specimens of urine should be cultured.

Tuberculin Skin Test

TST or Mantoux test is done by administering intradermally 0.1 mL of PPD of *MTB*, and the extent of induration caused by the injection is read after 48 to 72 hours. The test can be positive in active TB infection, LTBI, and those who have received BCG vaccination. This test is now rarely used.

Polymerase Chain Reaction

The sensitivities and specificities of polymerase chain reaction (PCR) on urinary samples are nearly 95% and 90%, respectively. A real-time quantitative PCR assay for amplifying *MTB* DNA and the gene for rifampicin resistance (GeneXpert *MTB*/RIF) is commercially available and is recommended by WHO. It is a rapid, affordable, near-point-of-care test for detecting *MTB* and rifampicin resistance simultaneously and can also be used for testing urine. An expanded new version called Xpert XDR allows for detection of resistance to isoniazid, injectable agents, and fluoroquinolones as well.¹⁸ GeneXpert *MTB*/RIF assays may give false-positive test results in treated patients because the *MTB* DNA lingers in tissues even after eradication of live mycobacteria. Novel, battery-operated, portable, single-module, near-patient/point-of-care versions of the GeneXpert system (GeneXpert Omni, GeneXpert Edge) are also available.

Whole-Blood Interferon- γ Release Assays

The test is based on quantifying the interferon (IFN)- γ released from the white blood cells that have been exposed to the mycobacterial antigens. Available as the QuantiFERON TB Gold and T Spot test, these assays can identify latent or active TB, replacing the traditional tuberculin test, and are reported to have a sensitivity and specificity of more than 90%.^{19,20} They can be performed in a single patient visit and are not altered by prior BCG vaccination.

Urine Tuberculosis Lipoarabinomannan Assay

Lipoarabinomannan forms part of the cell wall glycolipid of *MTB* bacterium and can be detected in the urine of patients with active or disseminated infection with kidney involvement. This immunochromatographic assay is recommended by the WHO for the diagnosis of HIV-associated TB in people with CD4⁺ lymphocyte counts of less than 200 cells/ μ L. The usefulness of this assay for diagnosing UG-TB has not yet been evaluated.²¹

Histopathology

Conventional needle biopsies are usually not done for diagnosis of kidney TB. However, ultrasound-guided fine-needle aspiration/biopsy from the lesion is useful in defining the granulomatous nature of the lesion. Histologic diagnosis is made by identifying the pathologic triad of caseating necrosis, loose aggregates of epithelioid histiocytes, and Langhans giant cells. Endoscopic examination (cystourethroscopy,

ureteroscopy, hysteroscopy, and laparoscopy) can be used for detecting anatomic abnormalities, localizing lesions, and obtaining biopsy tissue samples.

Imaging

All imaging findings may be normal in patients with early UG-TB, and findings are based on the presence of cavities, strictures, or calcifications. Imaging of the urinary tract helps to;

1. Localize the site of disease
2. Assess the extent of tissue destruction
3. Monitor response to treatment
4. Identify complications
5. Guide the needle to target sites for procuring samples for histologic, microbiological, and molecular analyses.

Plain Radiographs

Plain radiographs of the chest may show abnormal changes caused by active TB or healed TB in up to 50% of patients with UG-TB.²² In a plain radiograph of the abdomen, various types of calcification can be identified:

1. Amorphous, granular, or curvilinear calcification, within the kidney parenchyma: indicates early stages
2. Diffuse granular opacification or “cumulus cloud calcification”: often associated with active, granulomatous infection
3. Dense, punctate calcification: found in healed tuberculomas
4. Homogeneous, moderately dense, ground glass–like calcification: indicates calcified caseous tissue referred to as *putty kidney*
5. Triangular/ring-like calcifications: represents papillary necrosis
6. Lobar pattern with calcific rims outlining the periphery of distorted renal lobes: pathognomonic of TB.
7. Faint upper ureteral calcification along with any other kidney calcification: good marker of kidney TB.

Ultrasonography

In the early stages and in diffuse, infiltrative kidney TB, ultrasonography may appear normal. Granulomas are seen as small, hypoechoic intrarenal masses. In more extensive disease, changes similar to acute focal bacterial nephritis or chronic pyelonephritis may be seen. Large abscesses or tuberculomas distorting the kidney contour may be mistaken for tumors or cysts. Smaller parenchymal lesions, mucosal thickening, stenosis of the calyces, and ulcers can be identified by high-resolution ultrasonograms²³ (Fig. 54.8A–B). Various degrees of calcification, hydronephrosis, perinephric collections, bladder capacity, bladder emptying, and associated structural abnormalities in the genital tract can also be identified by ultrasonography.

Intravenous Urography

Intravenous urography (IVU) provides a simultaneous assessment of urinary tract anatomy and drainage. The important changes are:

1. Minimal calyceal dilation and loss of calyceal sharpness (early change) (see Fig. 54.6A)
2. Irregularity, fuzzy or feathery calyx (“moth-eaten” calyx)
3. Calyceal erosion and “phantom calyx”
4. Upward-pointing “hiked-up” pelvis
5. Kinking of the kidney pelvis (because of traction from a strictured infundibulum and parenchymal fibrosis): Kerr’s kink sign
6. Irregular ureter and areas of stricture and dilation: corkscrew ureter
7. Diffusely thickened ureters: pipestem ureter
8. Distal ureteric narrowing at ureterovesical junction
9. Small-capacity, thick-walled bladder with an elevated bladder base: thimble bladder (see Fig. 54.6B)

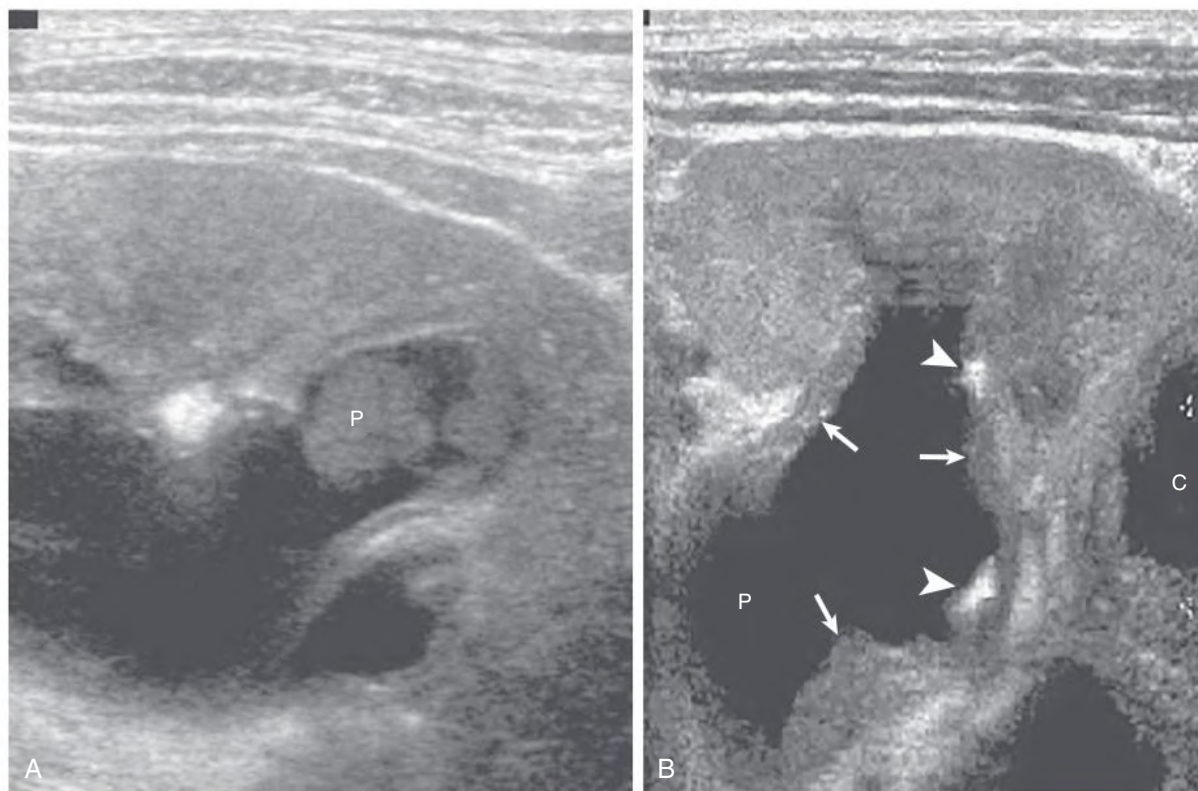


Fig. 54.8 High-Resolution Ultrasonography. (A) Sloughed, necrosed papilla (P) in the calyx. (B) Transverse ultrasound scan of kidney showing thickened mucosa (arrows) and calcification of calyces (C) and pelvis (P). There are calcifications of the wall of the calyx and pelvis (arrowheads). A parenchymal cavity (C) is also shown. (From Vijayaraghavan SB, Kandasamy SV, Arul M, et al. Spectrum of high resolution sonographic features of urinary tuberculosis. *J Ultrasound Med.* 2004;23[5]: 585–594.)

Computed Tomography Scan and Computed Tomography Urography

These imaging modalities are better than IVU and show abnormalities in greater detail. Computed tomography (CT) is a sensitive imaging modality to detect calcification, which is present in more than 50% of patients (see Fig. 54.5). Multidetector CT urography technology (MDCTU) using reformatted images such as multiplanar reconstruction and maximum intensity projection is replacing IVU for the assessment of kidney and urinary tract lesions.

Positron Emission Tomography–Computed Tomography Imaging

Positron emission tomography (PET) imaging using 2-deoxy-2-(fluorine-18)fluoro-d-glucose (^{18}F -FDG) provides functional information about sites with active inflammatory and immune cells that use glucose during metabolism. Anatomic and functional information is obtained by acquiring ^{18}F -FDG PET and CT data in a single scan. FDG PET-CT imaging does not help distinguish TB from cancer, thereby limiting its utility for specific diagnosis of UG-TB.

Magnetic Resonance Imaging

Magnetic resonance imaging (MRI) has a limited role because the sensitivity is low for detecting early lesions and abnormalities. For definitive diagnosis of tuberculosis, one major or two minor criteria are required.²⁴

Major criteria include:

- Granulomatous lesion in biopsy specimen
- AFB in urine or tissue by smear or culture
- Positive PCR

Minor criteria include:

- IVU/CT findings suggestive of GU-TB
- Hematuria
- Raised erythrocyte sedimentation rate (ESR)
- Pulmonary changes of old tuberculosis

LTBI is common and can become reactivated even decades later.

Currently, three tests are recommended by the WHO for detecting LTBI²⁵:

1. Tuberculin skin test (TST)
2. QuantiFERON-TB Gold
3. T-SPOT TB

Differential Diagnosis

UG TB mimics numerous diseases. Chronic nonspecific urinary tract infections may be confused with kidney TB, especially because secondary bacterial infection may complicate 20% of cases of kidney TB. Absence of response to usual antibiotics should raise suspicion of urinary TB. Conditions causing recurrent painless hematuria, such as IgA nephropathy and schistosomiasis, are often misdiagnosed as TB in endemic areas. In interstitial cystitis, lower urinary tract symptoms similar to tuberculous cystitis may occur, but the urinalysis does not show gross pyuria, and cultures for AFB are negative. On radiologic examination, chronic pyelonephritis, kidney papillary necrosis, medullary sponge kidney, calyceal diverticulum, renal carcinoma, xanthogranulomatous pyelonephritis, and multiple small kidney calculi need to be differentiated from TB. In a few reported cases of pseudotuberculous pyelonephritis, caseating granulomas resembling TB were observed in the kidney parenchyma, but no mycobacteria or other microorganisms were detected in the kidney tissue or urine culture.

MANAGEMENT OF UROGENITAL TUBERCULOSIS

The aims of management are to eradicate the infection, treat complications, and manage comorbidities/risk factors. The general precautions to prevent toxicity and adverse reactions in patients with CKD and dialysis are appropriate dose modification and administering drugs after hemodialysis. Antituberculous drugs form the mainstay of treatment (Table 54.3).

The latest recommendation by the WHO for treatment of drug-sensitive organisms is a 2-month intensive phase of quadruple therapy with first-line drugs (such as rifampicin, isoniazid, pyrazinamide, and ethambutol) daily, followed by a 4-month continuation phase of two drugs (rifampicin and isoniazid). The response in terms of resolution/improvement of systemic urinary symptoms and kidney function is assessed at 8 weeks. In immunosuppressed individuals, the continuation phase is extended to a total of 9 months. As with pulmonary TB, this regimen gives a cure rate of 90%.

In the case of drug-resistant UG-TB, there are scarce data and no clinical trials. As recommended by WHO for MDR pulmonary tuberculosis, an intensive 4- to 6-month course of therapy monitored by an experienced TB physician is necessary. The drug combinations may include five to six drugs initially as follows: gatifloxacin/moxifloxacin, ethionamide/prothionamide, kanamycin/amikacin, high-dose clofazimine/isoniazid with ethambutol, and pyrazinamide. This is followed by a 5-month continuation phase of clofazimine, ethambutol, pyrazinamide, and gatifloxacin or moxifloxacin.²⁶

In situations where the incidence of TB is high, LTBI is treated with isoniazid monotherapy for 6 months or rifampicin plus isoniazid daily for 3 months or rifapentine and isoniazid weekly for 3 months. In areas with lower incidence and for prevention, 6 to 9 months of isoniazid monotherapy or weekly rifapentine plus isoniazid for 3 months or rifampicin with or without isoniazid for 3 months is recommended. Newer drugs and repurposed drugs are used in MDR TB and XDR TB. Some drugs are still being examined in clinical trials (Table 54.4).

Surgical Treatment of Urogenital Tuberculosis

Surgical treatment as an adjunct to drugs is required in patients during or after TB drug therapy. Surgery is required for the following clinical settings:

1. Drainage for obstructed pelvicalyceal system
2. Drainage of abscesses
3. Reconstruction of ureters
 - Ureterocalicostomy
 - Reimplantation of ureters
 - Ileal replacement of ureter
4. Reconstructive surgery of the bladder (bladder augmentation)
5. Nephrectomy

Treatment Regimens in Special Situations

Women During Pregnancy and Lactation

Most antituberculous drugs are safe for use during pregnancy and lactation. However, streptomycin is ototoxic to the fetus and should be avoided. It is not necessary to isolate the baby from the mother unless the mother has active pulmonary tuberculosis. The baby should receive BCG immunization and isoniazid prophylaxis. Rifampin reduces the efficacy of oral contraceptives, and females should be advised to use alternative methods of contraception.

Patients With Liver Disease

In chronic liver disease, pyrazinamide is contraindicated. Isoniazid and two of the nonhepatotoxic drugs (streptomycin and ethambutol) can be used for 8 to 12 months. If rifampin is used, liver function

should be closely monitored. In those with acute hepatitis unrelated to TB or its therapy, it would be safer to defer chemotherapy until the acute hepatitis has resolved. If immediate treatment of TB during acute hepatitis is mandatory, streptomycin plus ethambutol for 3 months followed by isoniazid and rifampin for 6 months is advised. Excessive alcohol consumption should be avoided.

Patients With Chronic Kidney Disease

Because streptomycin and ethambutol are excreted by the kidney, dosage modification of these drugs is necessary when GFR is less than 60 mL/min/1.73 m². For ethambutol, the full dose is administered every 24 to 36 hours if GFR is 10 to 50 mL/min/1.73 m² and every 48 hours if GFR is less than 10 mL/min/1.73 m². Monthly questioning for symptoms of visual dysfunction (alterations in visual fields, acuity, blue-green vision) and periodic ophthalmic examination may identify early ethambutol toxicity, at the stage of potential reversibility. Streptomycin is used only rarely now because of ototoxicity. It is avoided in older individuals, and the dose is 15 mg/kg every 48 to 72 hours if GFR is 10 to 50 mL/min/1.73 m² and every 72 to 96 hours if GFR is less than 10 mL/min/1.73 m². In patients with CKD, isoniazid, rifampin, and pyrazinamide, which are eliminated by the biliary route, can be given in normal dosages. In patients on dialysis, it is ideal to administer these drugs under supervision after each dialysis session.

Kidney Allograft Recipients

No clear-cut recommendations exist for GU-TB in native kidneys after transplantation. It is preferable to complete at least 6 weeks of therapy for all types of TB and reassessment before considering transplantation. Ideally, the patient could be considered for kidney transplantation after 6 months of treatment. Rifampin is avoided in patients receiving a calcineurin inhibitor (CNI) because enzyme induction may make maintenance of adequate CNI blood levels difficult. For patients with a kidney allograft, a suitable treatment regimen is with isoniazid and ethambutol for 18 months (with dose adjusted for GFR), combined with ofloxacin 200 mg twice daily for the first 9 months and pyrazinamide 750 mg twice daily for the first 3 months.²⁷ If rifampin is used in those receiving a non-CNI-based immunosuppressive regimen, the maintenance dose of prednisolone should be doubled.

Human Immunodeficiency Virus and Tuberculosis

In patients with HIV who are already on antiretroviral treatment (ART), conventional 6-month treatment with 4 drugs for the first 2 months followed by 2 drugs for the remaining 4 months is sufficient. In patients with HIV infection who are not on ART, antituberculous drugs are continued for 9 months. If HIV and TB are diagnosed simultaneously, antituberculous drugs are started first, followed by ART after 15 days if the CD4 count is less than 50 cells/ μ L and after 30 days if the CD4 cell count is less than 100 cells/ μ L.

Patients Who Do Not Respond to Treatment

Failure of clinical or radiologic improvement with treatment may signify poor compliance, inadequate regimen and/or dosage, incorrect diagnosis, delayed response, MDR TB, XDR TB, or the paradoxical reaction known as *immune reconstitution inflammatory syndrome*. This syndrome is characterized by unexpected worsening of symptoms or appearance of new lesions, including lymphadenopathy, serosal effusions, and pleural infiltrates. This syndrome is more common in those receiving ART for coexisting HIV and TB infections.²⁸

Newer Therapies

The recent introduction of two new drugs, bedaquiline and delamanid, for rifampicin-resistant tuberculosis and ongoing studies on

TABLE 54.3 Antituberculous Drugs: Dosage, Actions, and Side Effects^a

Drug	Dose Form	Dosage	Mode of Action	Dose Modification GFR (50–10 mL/min)	Dose Modification GFR (<10 mL/min)	Side Effects	Remarks
Isoniazid (INH) ^b	Tablet: 100, 300 mg	PO: 5 mg/kg/day (max 300 mg/day)	Bactericidal for groups I and II Interferes with mycolic acid synthesis	Nil	66%	Hypersensitivity Peripheral neuritis Hepatitis	Administer pyridoxine 50 mg/day
Rifampin (rifampicin) ^b	Tablet/capsule: 150, 300, 450 mg	PO: 10 mg/kg/day (max 600 mg/day)	Bactericidal for groups I, II, and III Interferes with protein synthesis by inhibiting RNA polymerase	Nil	Nil	Febrile reactions Acute interstitial nephritis Hepatitis	Dose adjustment when using calcineurin inhibitors or oral contraceptives May be used in pregnancy and lactation
Pyrazinamide ^b	Tablet: 400, 500 mg	PO: 25 mg/kg/day (max: 2 g/day)	Bactericidal for semidormant <i>Mycoplasm tuberculosis</i>	Nil	50% ^c	Hyperuricemia Hepatotoxicity Photosensitivity	Primary hepatic metabolism Monitor liver functions and uric acid
Ethambutol	Tablet: 100, 400 mg	PO: 15–25 mg/kg (max 2.5 g/day)	Bacteriostatic Bactericidal in high concentrations Inhibits cell wall synthesis	75% ^c	Avoid or 50% ^c	Optic neuritis (monitor for color vision)	Primarily renal excretion May be used during pregnancy and lactation
Streptomycin	Injection: 1, 0.75 g	IM: 15–25 mg/kg/day (max 1 g) Age >60 y: 75% dose	Bactericidal Inhibit protein synthesis	50% ^c	25% ^c	Ototoxicity Vestibulotoxicity Hypokalemia Hypomagnesemia	Therapeutic drug monitoring advised Peak level for efficacy Trough level for toxicity
Amikacin	IM/IV: ampoules, 500, 1000 mg	15 mg/kg/day (max 1 g/day)	Bactericidal Inhibit protein synthesis	50% ^c	25% ^c	Ototoxicity Vestibulotoxicity Hypokalemia Hypomagnesemia	Cleared by hemodialysis Supplement dose after HD or give regular full dose after dialysis Avoid coadministration of loop diuretics TDM recommended
Levofloxacin	PO: tablet, 250, 500, 750 mg IV: ampoules, 500–750 mg	750–1000 mg/day	Bactericidal Inhibit protein synthesis—DNA gyrase	50% ^c (500–750 mg q36h)	25% ^c (250–500 mg q48h)	Severe upper GI symptoms Prolonged QTc Arthralgia Tendon rupture	Hepatic and renal elimination Used in liver disease More effective than ciprofloxacin and ofloxacin
Moxifloxacin	PO: Tablet, 400 mg IV: infusion, 400 mg in 100 mL	400 mg/day	Bactericidal	Nil	Nil	Severe upper GI symptoms Prolonged QTc Arthralgia Tendon rupture	Not removed by HD
Capreomycin	IM/IV: 1-g vial	15 g/kg/day (max 1 g/day) on 5–7 days/wk	Strongly bactericidal Inhibits protein synthesis	75%	25% ^c	Nephrotoxicity Ototoxicity Hypokalemia Hypomagnesemia Hepatotoxicity when combined with other anti-TB drugs	Used only for MDR TB

TABLE 54.3 Antituberculous Drugs: Dosage, Actions, and Side Effects^a—cont'd

Drug	Dose Form	Dosage	Mode of Action	Dose Modification GFR (50–10 mL/min)	Dose Modification GFR (<10 mL/min)	Side Effects	Remarks
Ethionamide	PO	15–25 mg/kg/day (max 1 g/day in 2 divided doses)	Weakly bactericidal Blocks mycolic acid synthesis	Nil	50%	Hepatotoxic Severe GI symptoms Hypothyroidism Gynecomastia	Not to be combined with cycloserine
Cycloserine	PO	15–25 mg/kg/day	Bacteriostatic	50% ^c	25% ^c	Neurotoxicity Seizures Peripheral neuropathy	Maintain peak level <35 µg/mL Cleared by dialysis Avoid in pregnancy and lactation

^aThe main drugs are listed with dosage form, dosage, mode of action, side effects, and dose modifications in those with low glomerular filtration rate. *M. tuberculosis* exists as three subpopulations. Group I is extracellular, occurs mainly in cavitating lesions, and responds to streptomycin, isoniazid, and rifampin. Group II resides intracellularly in macrophages, replicates slowly, and responds to pyrazinamide, isoniazid, or rifampin. Group III organisms exist within closed caseous lesions, survive better in neutral pH, replicate slowly, and respond best to rifampin.

^bPrescribed dose to be given after hemodialysis.

^cDose reduction and/or increasing dosing intervals.

GFR, Glomerular filtration rate; GI, gastrointestinal; HD, hemodialysis; IM, intramuscular; IV, intravenous; MDR TB, multidrug-resistant TB; PO, orally; QTc, corrected QT interval on electrocardiogram; TB, tuberculosis; TDM, therapeutic drug monitoring.

TABLE 54.4 Newer Drugs: Clinical Pharmacology

Name of Drug	Class/Mechanism of Action	Dose/Route of Administration	Side Effects	Remarks
Bedaquiline	Diarylquinoline Inhibits mycobacterial ATP synthase Core drug in rifampicin resistant tuberculosis All-oral regimen Capable of killing both actively replicating and nonreplicating persistent mycobacteria in cells	Oral/400 mg once daily for 2 wk, followed by 200 mg three times/wk for 24 wk (total 26 wk)	QT prolongation Hepatitis GI toxicity	Recommended by WHO for treatment of MDR TB May be used in pregnancy
Delamanid	Nitroimidazole Inhibits mycolic acid synthesis	Oral/100 mg twice daily for 24 wk	QT prolongation GI toxicity	Recommended by WHO Drug of choice for children <6 y with rifampicin-resistant tuberculosis Avoid in pregnancy
Pretomanid	Nitroimidazole	Oral 200 mg once daily	Peripheral neuropathy (81%) Headache (28%) Skin rash (21%)	Experimental No human trials yet
Clofazimine	Active against inactive nonreplicating mycobacterium	Oral 100–200 mg once daily	Red discoloration of skin, eyes, body fluids Photosensitivity GI toxicity	Avoid in pregnancy
Linezolid/sutezolid	Oxazolidinone/antibiotic	Oral or IV 600 mg once daily or 1200 mg once daily for 14 d	Myelosuppression GI toxicity Optic neuritis Peripheral neuropathy	Avoid in pregnancy
Telacebec	Imidazopyridine (IPA) compound Mycobacterial respiratory chain inhibitor Targets MTB cellular energy production by inhibiting (with high specificity) cytochrome bc1 complex	Oral	—	Completed phase II trial
Delpazolid	Oxazolidinone Inhibits protein synthesis Binding to domain V of rRNA			Ongoing phase II trials

GI, Gastrointestinal; MDR TB, multidrug-resistant tuberculosis; MTB, *Mycobacterium tuberculosis*; WHO, World Health Organization.

rifapentine, clofazimine, and pretomanid may alter the treatment landscape in the future (see Table 54.4).²⁹ There is growing interest in the use of repurposed drugs such as linezolid and clofazimine. The second novel drug, delamanid, is recommended by WHO as the drug of choice for treating children younger than 6 years with rifampicin-resistant TB.³⁰

A single interstitial injection of autologous bone marrow–derived mesenchymal stem cells (MSCs) into the bladder showed considerable reduction of inflammation as well as a reduction in the development of fibrosis and bladder wall deformity in animal models.³¹

SELF-ASSESSMENT QUESTIONS

- All of the following are features of immune reconstitution inflammatory syndrome (IRIS) *except*:
 - Development of new lesions
 - Worsening of symptoms
 - Severity inversely proportional to the load of organisms
 - Common in MDR TB
 - Common in patients with simultaneous treatment of coexisting HIV infection
- The following antituberculous drugs act as bactericidal agents when used in therapeutic doses *except*:
 - Rifampicin
 - Isoniazid
 - Amikacin
 - Ethambutol
 - Pyrazinamide
- The following symptoms point to suspicion of kidney tuberculosis *except*:
 - Recurrent painless macroscopic hematuria
 - Acid, sterile pyuria
 - Sterility in women/epididymitis in men
 - Lack of constitutional symptoms
 - Icterus
- Which of the following antituberculous drug combinations require dose modifications in patients with kidney failure?
 - Isoniazid and rifampin
 - Streptomycin and ethambutol
 - Pyrazinamide and rifampin
 - Isoniazid and pyrazinamide
- Which drug is uniformly effective against all three groups of *Mycobacterium tuberculosis*—group I (extracellular), group II (intracellular), and group III (closed caseous lesions)?
 - Isoniazid
 - Rifampicin
 - Pyrazinamide
 - Ethambutol
 - Streptomycin

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